higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

POWER MACHINES N5

11 APRIL 2019

This marking guideline consists of 13 pages.

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QUESTION 1

1.1 Clausius's theory states that heat can flow from a hot substance (higher temperature) to a cold substance (lower temperature) without the aid of external work (naturally) and without naturally reversing the flow process. (2)

1.2
- To control or maintain the desired speed of an engine or a mechanical mechanism.
- To regulate or keep the speed of an engine within required limits or specifications. (Any 1 x 2) (2)

1.3
- Carbon dioxide or CO_2
- Water vapour (steam) or H_2O (2)

1.4
- The name of a marker
- The factory number
- The year of construction
- The design pressure in Pascal (Any 2 x 1) (2)

1.5
- Impulse turbine
- Reaction turbine (2)

[10]

QUESTION 2

2.1

$$from, \frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} = mR$$

$$P_1V_1 = P_2V_2 ... T_1 = T_2 \surd$$

$$V_2 = \frac{P_1V_1}{P_2} \surd$$

$$= \frac{220 \times 0,5}{1500} \checkmark$$

$$\therefore V_2 = 0,073m^3 \checkmark$$

Alternative method :

$$from, P_1V_1 = mRT_1$$

$$m = \frac{P_1V_1}{RT_1} \surd$$

$$= \frac{220 \times 0,5}{0,296 \times 293} \surd$$

$$\therefore m = 1,26833kg \surd$$

$$V_2 = \frac{mRT_2}{P_2} \surd$$

$$= \frac{1,26833 \times 0,296 \times 293}{1500} \surd$$

$$\therefore V_2 = 0,073m^3 \surd$$

(3)

2.2

$$from, \frac{P_2V_2}{T_2} = \frac{P_3V_3}{T_3} = mR$$

$$\frac{P_2}{T_2} = \frac{P_3}{T_3} ... V_2 = V_3 \surd$$

$$P_3 = \frac{P_2T_3}{T_2} \surd$$

$$= \frac{1500 \times 293}{303} \checkmark$$

$$\therefore P_3 = 1450,495kPa \checkmark$$

(3)

2.3

$$R = C_p - C_v$$

$$C_p = R + C_v \surd$$

$$C_p = 0,296 + 0,709 \surd \surd$$

$$C_p = 1,005\,kJ/kg.K \surd$$

(2)

2.4

isothermal process:

$$Q_{isothermal} = P_1V_1 \ln \frac{V_2}{V_1} + mC_v(T_2 - T_1)$$

$$= (220 \times 0,5) \surd \ln \frac{0,073}{0,5} + 0 \surd$$

$$Q_{isothermal} = -211,656kJ \checkmark$$

cons tant volume process:

$$Q_{vol} = P\Delta V + mc_V(T_3 - T_2)$$

$$= \surd 0 + 0,709 \surd (303 - 293) \surd$$

$$Q_{vol} = 7,09\ kJ/kg \checkmark$$

(4)

2.5

for isothermal process:

$$\Delta U_{isothermal} = mC_v(T_2 - T_1)$$

$$= 0,709(293 - 293) \surd$$

$$\Delta U = 0kJ/kg \checkmark$$

for cons tant volume:

$$\Delta U_{volume} = mC_v(T_3 - T_2)$$

$$= 0,709(303 - 293) \surd$$

$$\therefore \Delta U_{volume} = 7,09kJ/kg \checkmark$$

$$\Delta U_{total} = \Delta U_{volume} + \Delta U_{isothermal}$$

$$= 0 + 7,09 \surd$$

$$= 7,09kJ/kg \surd$$

(4)

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2.6

$$\Delta S_{total} = \left[R \ln \frac{V_2}{V_1} + C_V \ln \frac{T_2}{T_1} \right] + \left[C_P \ln \frac{T_3}{T_2} - R \ln \frac{P_2}{P_3} \right]$$

$$= \left[0,296 \ln \left(\frac{0,073}{0,5} \right) + 0,709 \ln \left(\frac{293}{293} \right) \right] + \left[1,005 \ln \left(\frac{303}{293} \right) - 0,296 \ln \left(\frac{1500}{1450,495} \right) \right]$$

$$= (-0,5695) + (0,033728 - 0,00993)$$

$$\therefore \Delta S_{total} = -0,546 kJ / kg$$

Alternative method :

$$\Delta S_{total} = \left[R \ln \frac{V_2}{V_1} + C_V \ln \frac{T_2}{T_1} \right] + \left[R \ln \frac{V_3}{V_2} + C_V \ln \frac{T_3}{T_2} \right]$$

$$= \left[0,296 \ln \left(\frac{0,073}{0,5} \right) + 0,709 \ln \left(\frac{293}{293} \right) \right] + \left[0,296 \ln \left(\frac{0,073}{0,073} \right) + 0,709 \ln \left(\frac{303}{293} \right) \right]$$

$$= (-0,56955) + (0,02379)$$

$$\therefore \Delta S_{total} = -0,546 kJ / kg$$

(4)
[20]

QUESTION 3

3.1

$At\ P_1 = 2500 kPa$:

$t_s = 223,9^o C$

$t_{ss} = 250^o C$

$h_g = 2801 kJ/kg$

$$h_1 = m \left[h_g + c_p \left(t_{ss} - t_s \right) \right]$$

$$= 2 \left[2801 + 2 \left(250 - 223,9 \right) \right]$$

$$= 2 \times 2853,2$$

$$= 5706,4\ kJ / kg$$

(4)

3.2

$At\ P_1 = 2500 kPa$:

$h_1 = 2853,2\ kJ / kg$

$$v_1 = \frac{n-1}{n} \left(\frac{h_{su} - 1941}{P} \right)$$

$$= \frac{1,3-1}{1,3} \left(\frac{2853,2 - 1941}{2500} \right)$$

$$= 0,0842 \frac{m^3}{kg} \times 2kg$$

$$= 0,1684 m^3$$

(4)

3.3 $At\ P_2 = 250kPa:$

$h_f = 535kJ/kg$ ✓

$h_{fg} = 2181kJ/kg$ ✓

$h_2 = m\left[h_f + xh_{fg}\right]$

$= 2\left[535 + 0,91(2181)\right]$

$= 2 \times 2519,71$ ✓

$= 5039,42\,kJ/kg$ √ (4)

3.4 $At\ P_2 = 250kPa:$

$V_g = 0,7184m^3/kg$ ✓

$V_2 = xV_g$

$= 0,91 \times 0,7184$

$= 0,6537\frac{m^3}{kg} \times 2\ kg$

$= 1,307\ m^3/kg$ (4)

3.5 $U_1 = h_1 - p_1V_1$

$= 5706,4 - (2500 \times 0,1684)$

$= 5285,4\ kJ$

$U_2 = h_2 - p_2V_2$

$= 5039,42 - (250 \times 1,307)$

$= 4712,67\,kJ$

$\Delta U = (U_1 - U_2)$

$= 5039,42 - 4712,67$

$= 572,85\ kJ\ (gained)$

OR

$\Delta U = U_2 - U_1$

$= 4712,67 - 5039,42$

$= -572,85$

$= 572,85\ kJ\ (lost)$

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Alternative Method

$U_1 = h_1 - p_1V_1$

$= 2853,2 - (2500 \times 0,0842)$

$= 2642,7kJ / kg$

$U_2 = h_2 - p_2V_2$

$= 2519,71 - (250 \times 0,6537)$

$= 2356,285kJ / kg$

$\Delta U = m(U_1 - U_2)$

$= 2(2642,7 - 2356,285)$

$= 572,83kJ$

$= 572,83\ kJ(gained)$

OR

$\Delta U = m(U_2 - U_1)$

$= 2\ (\ 2356,285 - 2642,7\ )$

$= -572,83\,kJ$

$= 572,83\ kJ(lost)$ (4)

[20]

QUESTION 4

4.1 4.1.1 $P = 11\ kPa$

From the steam table at :

$P = 11\ kPa \quad h_f = 200\ kJ / kg$ ✓

$h_{fg} = 2388\ kJ / kg$ ✓

$t_c = 41,6\ ^oC \quad h_c = 174\ kJ / kg$ ✓

heat emitted by steam = heat absorbed by wate

$m_s\left[h_f + xh_{fg} - h_c\right] = mc_p\Delta T$

$\frac{3200}{3600}\left[200 + (0,91 \times 2388) - 174\right] = 12 \times 4,187 \times (T_2 - 20)$ ✓

$T_2 = \frac{1954,738}{50,244} + 20$ ✓

$\therefore T_2 = 58,905\ ^oC$ (6)

4.1.2 *From the steam table at* $t_c = 41,6^oC$

$P_s = 8kPa$ √

$P_T = P_a + P_s$ √

$P_a = 11 - 8$ √

$P_a = 3\ kPa$ √

$m_a = \frac{1,5\ kg\ air}{1000\ kg\ steam} \times 3200\ kg\ steam$ √

$m_a = 4,8\ kg\ air / hr$ √

$P_a V_a = mRT_a$ √

$V_a = \frac{mRT_a}{P_a}$ √

$V_a = \frac{4,8 \times 0,296 \times 314,6}{60 \times 3}$ √

$V_a = 2,483\ m^3 / \min$ √ (5)

4.2.1

Symbol	%	*Equation*	O_2	CO_2	H_2O	N_2
H_2	37	$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$	$\frac{1}{2} \times 0,37 = 0,185$	0	$1 \times 0,37 = 0,37$	0
CH_4	29	$CH_4 + 2O_2 \rightarrow CO_2 + 2\ H_2O$ √	$2 \times 0,29 = 0,58$	$1 \times 0,29 = 0,29$	$2 \times 0,29 = 0,58$	0
CO	18	$CO + \frac{1}{2}O_2 \rightarrow CO_2$	$\frac{1}{2} \times 0,18 = 0,09$	$1 \times 0,18 = 0,18$	0	0
CO_2	1		0	0,01	0	0
N_2	15	N_2	0	0	0	$0,15 + 4,701 \times \frac{79}{100}$
Total	100		0,855	0,48	0,95	3,366

(4)

$$V_a = \frac{100}{21} \times 0,855$$

$$V_a = 4,071 \ \frac{m^3 \ air}{m^3 \ fuel}$$ (1)

OR

Alternative method:

$$From, V_{o_2}(required) = V_{H_2} + V_{CH_4} + V_{CO}$$

$$From, C_xH_y + \left(x + \frac{y}{4}\right)O_2 = xCO_2 + \left(\frac{y}{2}\right)H_2O$$

$$CH_4 + \left(1 + \frac{4}{4}\right)O_2 = CO_2 + \left(\frac{4}{2}\right)H_2O \quad \surd \surd \surd$$

$$CH_4 + 2O_2 = CO_2 + 2H_2O \quad \surd$$

$$Now, V_{o_2}(required) = (0,37 \times 0,5) + (0,29 \times 2) + (0,18 \times 0,5) \quad \surd \surd \surd$$

$$\therefore V_{o_2} = 0,855 m^3_{o_2} / m^3_{fuel} \quad \surd$$

$$as, V_{o_2} = 23\% \ of \ V_{air}$$

$$now, \ V_{air} = \frac{0,855}{0,21} \quad \surd$$

$$\therefore V_{air} = 4,071 m^3_{air} / m^3_{fuel} \quad \surd$$ (5)

4.2.2 Pr*oducts*:

$V_{H_2O} = 0,48 \ m^3 \ H_2O / m^3 \ fuel$

Percentage:

$$V_{H_2O} = \frac{0,48}{4,795} = 10 \ \% \quad \surd$$

$V_{CO_2} = 0,95 \ m^3 \ CO_2 ¿ m^3 \ fuel$

$$V_{CO_2} = \frac{0,95}{4,795} = 19,8 \ \% \quad \surd$$

$V_{N_2} = 3,366 \ m^3 \ N_2 ¿ m^3 \ fuel$

$$V_{N_2} = \frac{3,366}{4,795} = 70,2 \% \quad \surd$$

$Total = 4,795 \ m^3 \ products / m^3 \ fuel$ $Total = 100 \%$ $\surd$ (4)

Alternative method :

$$\sum V_{comb} = \sum V_{CO_2} + \sum V_{H_2O} + \sum N_2$$

$$= \left[(0,29 \times 1) + (0,18 \times 1) + 0,01\right] + \left[(0,37 \times 1) + (0,29 \times 2)\right] + \left[(0,79 \times 4,701) + 0,15\right]$$

$$\therefore \sum V_{comb} = 5,29379 m^3 / m^3_{fuel}$$

$$\%\sum V_{CO_2} = \frac{0,48}{5,29379} \times 100\% = 9,067\%$$

$$\%\sum V_{H_2O} = \frac{0,95}{5,29379} \times 100\% = 17,946\%$$

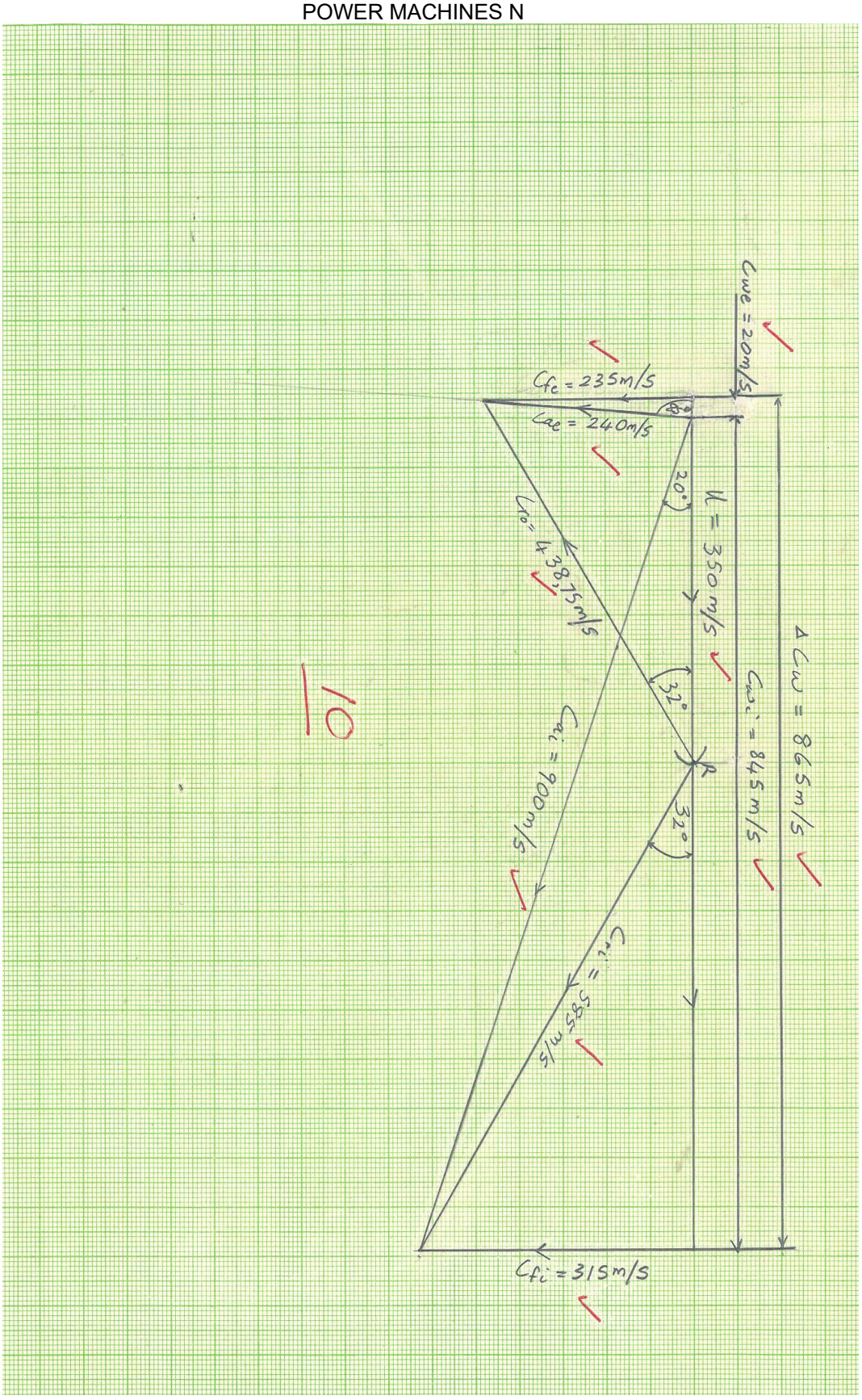
$$\%\sum N_2 = \frac{3,86379}{5,29379} \times 100\% = 72,987\%$$

$$\%\sum V_{comb} = 9,067 + 17,946 + 72,987 = 100\%$$

[20]

QUESTION 5

5.1


Cwe = 20m/s
Cfe = 235m/s
Cae = 240m/s
20°
U = 350 m/s
Cro = 438,75m/s
32°
Cri = 800m/s
Cwi = 845 m/s
ΔCw = 865m/s
32°
Cri = 585m/s
Cfi = 315m/s
10

(10)

5.2 5.2.1 *From the velocity diagram*:

$\beta = 85^0$ ✓ (1)

5.2.2 *From the velocity diagram*:

$\alpha = 32^0$ ✓ (1)

5.3 5.3.1 *From the velocity diagram*:

$\Delta C_w = 20 + 845$ √

$= 865 m/s$ √

$\overset{o}{m} = \frac{2}{60} = 0,03333 m/s$ √

$P_{turbine} = \overset{o}{m} \Delta C_w \times U$

$= 0,03333 \times 865 \times 350$ √

$\therefore P_{turbine} = 10,091 kW$ ✓ (3)

[15]

QUESTION 6

6.1 $V_s = V_1 - V_3 \ldots\ldots(1)$ √

$and\ V_c = V_3$ √

$\therefore\ V_s = V_1 - 0,05\,V_s$ √

$V_1 = 1,05\,V_s \ldots\ldots\ldots\ldots(2)$ √

$V_3 = \frac{5}{100} \times V_s$ √

$V_3 = 0,05\,V_s \ldots\ldots(3)$ √

$V_{4,} = V_3 \left(\frac{P_4}{P_3}\right)^{\frac{1}{n}}$

$V_4 = (0,05 V_S)\left(\frac{700}{100}\right)^{\frac{1}{1,3}}$ √

$V_4 = 0,2233\,V_s \ldots\ldots\ldots(4)$ √

$V_e = \frac{15m^3}{\min} \times \frac{\min}{350r}$ √

$V_e = 0,0428\ m^3$ √

$V_e = V_1 - V_4$

$0,0428 = 1,05\,V_s - 0,2233\,V_s$ ✓

$= 0,8267 V_s$ √

$V_s = \frac{0,0428}{0,8267}$ √

$V_s = 0,05177\ m^3$ √

$V_1 = 1,05 \times 0,05177$ ✓

$\therefore V_1 = 0,0543\ m^3$ √

Alternative method :

$\text{Since}, V_1 = SV_s + CV$

$also, V_1 = SV_e + V_4$

$$from, \eta_{vol} = \left\{1 - \frac{CV}{SV_s}\left[\left(\frac{P_2}{P_1}\right)^{\frac{1}{n}} - 1\right]\right\} \times 100\%$$

$$= \left\{1 - \frac{\sqrt{0,05SV_s}}{SV_s}\left[\left(\frac{\sqrt{700}}{100}\right)^{\frac{1}{1,3}} - 1\right]\right\} \times 100\%$$

$$= \left\{1 - 0,05\left[4,46761 - 1\right]\right\} \times 100\% \checkmark$$

$$= 82,66195\% \checkmark$$

$$also, \eta_{vol} = \frac{\overset{0}{SV_e}}{\overset{0}{SV_s}} \times 100\%$$

$$\overset{0}{SV_s} = \frac{\overset{0}{SV_e}}{\eta_{vol}}$$

$$= \frac{15}{0,8266195} \checkmark$$

$$\therefore \overset{0}{SV_s} = 18,146197 m^3 / \min \checkmark$$

$$also, \overset{0}{SV_s} = SV_s \times N$$

$$SV_s = \frac{18,146197}{350} \checkmark$$

$$= 0,05185 m^3 \checkmark$$

$$thus, V_1 = SV_s + CV$$

$$= 0,05185 + (0,05185 \times 0,05)$$

$$\therefore V_1 = 0,054 m^3 / \min \sqrt{}$$

(9)

6.2

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}}$$

$$T_2 = T_1\left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} \sqrt{}$$

$$= (315)\left(\frac{700}{100}\right)^{\frac{1,3-1}{1,3}} \sqrt{}$$

$$\therefore T_2 = 493,552\ K \checkmark$$

(2)

6.3

$$Power_{comp} = \frac{n}{n-1} P_1 \overset{o}{V}_e \left[\left(\frac{P_2}{P_1} \right)^{\frac{n-1}{n}} - 1 \right]$$

$$= \frac{1,3 \times 100 \times 15}{(1,3-1) \times 60} \left[\left(\frac{700}{100} \right)^{\frac{1,3-1}{1,3}} - 1 \right]$$

$$= 61,40688kW$$

$$from, \eta_{mech} = \frac{Power_{comp}}{Power_{motor}} \times 100\%$$

$$thus, Power_{motor} = \frac{61,40688}{0,85}$$

$$\therefore Power_{motor} = 72,243kW$$

(4)
[15]

TOTAL: 100

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